

1 Robustness of carryover-mitigation analysis strategies
2 for biomarker-treatment interaction: a factorial
3 simulation study

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1 Abstract

Background. Carryover effects in aggregated N-of-1 and crossover trials degrade the detectability of biomarker-treatment interactions. Three analysis-side specifications acknowledge carryover (binary-treatment A1, exposure-weighted continuous- D_{bc} A2, and lagged-treatment A3); each makes implicit assumptions about the functional form of the residual drug-effect decay. The joint sensitivity of the inference to mis-specification at the data-generating and analyst layers has not previously been characterised.

Methods. We conduct an ADEMP-compliant factorial simulation study¹ crossing three data-generating carryover decay forms (linear, exponential, Weibull) against the three analysis-model specifications, under both mean-moderation (Architecture~A) and multivariate-normal differential-correlation (Architecture~B) biomarker data-generating processes. The principal factorial covers 540 cells at 500 replicates per cell. Five sensitivity blocks (S1-S5) vary autocorrelation strength, analyst-versus-DGP half-life mismatch, dropout rate, between-subject heterogeneity, and biomarker correlation strength. All performance measures are reported with Monte Carlo standard errors per Morris et al. (2019).

Results. Across the principal factorial and all sensitivity blocks, the exposure-weighted A2 specification consistently produced the highest power, with an absolute advantage of approximately 10 percentage points over A1 at the reference cell (Architecture~B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ week; A2 = 0.860, A1 = 0.766, A3 = 0.770; approximately five MCSE of separation). The ranking $A2 > A1 \approx A3$ was preserved across designs, architectures, sample sizes, and DGP carryover half-lives. A2 retained its advantage when the analyst-assumed half-life departed from the true half-life by a factor of two or four. A high-precision rerun at $N \in \{35, 70\}$ and $t_{1/2}^{DGP} \in \{0.5, 1.0\}$ confirmed the ranking at MCSE 0.020 with a paired McNemar test $p = 6.6 \times 10^{-17}$ at the highest-leverage cell.

Conclusions. A2 is the recommended default for the analysis of biomarker-

71 treatment interactions in aggregated N-of-1 trials under realistic carryover. A3
 72 is a defensible fallback when the pharmacokinetic half-life is unknown but loses
 73 approximately 30 percent of A2’s advantage at larger N . A1 (the strict-RM-
 74 ANOVA analogue) is dominated across the entire grid.

75 **2 Introduction**

76 Biomarker-moderated treatment effects in within-subject designs depend on how
 77 residual drug effect is modelled in both the data-generating process (DGP) and the
 78 analysis model. The DGP architecture comparison in the companion manuscript
 79 at `analysis/report/01-dgp-mean-moderation-vs-mvn/` establishes that carry-
 80 over has qualitatively different power consequences under the two architectures².
 81 That prior work held the decay form fixed at exponential and evaluated a single
 82 analysis model.

83 This paper addresses a separate question: given that the trial designer will spec-
 84 ify a carryover decay form (explicitly or implicitly) and the analyst will choose
 85 an analysis-model carryover specification, how sensitive is the realised power to
 86 mis-specification at each layer? The practical motivation is that the true pharma-
 87 cokinetic decay of residual drug effect is rarely known with precision at the design
 88 stage, and the analyst typically defaults to a convenient parametric form.

89 We perform a factorial simulation study crossing three DGP decay forms (linear,
 90 exponential, Weibull) with three analysis-model carryover specifications (binary-
 91 treatment, exposure-weighted, lagged-treatment). We run the full grid under both
 92 DGP architectures to establish whether robustness rankings transfer across archi-
 93 tectures. The study is designed and reported following the ADEMP framework
 94 of Morris, White, and Crowther¹, which structures simulation studies around five
 95 elements: aims, data-generating mechanisms, estimands, methods, and perfor-
 96 mance measures. Each performance measure is reported per cell with its Monte
 97 Carlo standard error.

98 **3 Simulation design**

99 This section organises the study under the ADEMP labels of Morris et al.¹. §3.1
 100 defines the data-generating mechanisms, §3.2 the estimands, §3.3 the methods
 101 (analysis-model specifications), §3.4 the factorial grid, §3.5 the parameter settings,
 102 and §3.6 the performance measures. Reproducibility metadata (R version, key
 103 package versions, the RNG seed strategy, and elapsed time) are recorded in §3.7
 104 (Computing environment).

105 **3.1 Aims**

106 The study quantifies the cost of analysis-model carryover mis-specification in ag-
 107 gregated N-of-1 and crossover trials, and identifies which mitigation strategy is
 108 most robust across the DGP decay-form grid. A secondary aim is to verify that the

robustness rankings transfer across data-generating-process architectures (mean moderation versus differential correlation).

3.2 Data-generating mechanisms

Both DGP architectures from the companion manuscript are retained:

- **Architecture A (direct mean moderation).** Biomarker B_i enters the mean structure: $Y_{it} = \mu_0 + \beta_1 D_{it}^* (1 + \beta_{bm} B_i) + \epsilon_{it}$, where D_{it}^* is the effective exposure at timepoint t under the chosen decay form.
- **Architecture B (MVN differential correlation).** Biomarker and biological response are jointly drawn from a multivariate normal with treatment-state-dependent correlation $\text{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form.

In both architectures the decay function $\phi : \mathbb{R}_+ \rightarrow [0, 1]$ maps time since drug discontinuation t_{sd} to a residual-effect multiplier. We evaluate three forms parameterised to match a common half-life $t_{1/2}$:

3.2.1 Linear decay

$$\phi_{\text{lin}}(t_{sd}) = \max\left(0, 1 - \frac{t_{sd}}{2t_{1/2}}\right)$$

Residual effect vanishes completely at $t_{sd} = 2t_{1/2}$. This form is biologically implausible for drugs with first-order elimination kinetics but represents a defensible pragmatic assumption when only the “carryover has fully worn off by time X” information is available.

3.2.2 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp(-\lambda t_{sd}), \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

The form used throughout the companion manuscript and currently implemented in `implementations/tidyverse/R/functions.R`. Corresponds to first-order elimination kinetics and is the default assumption in classical pharmacokinetics.

3.2.3 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-(\lambda_w t_{sd})^k\right), \quad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k controls whether the tail is heavier ($k < 1$, slow elimination) or lighter ($k > 1$, accelerated washout) than exponential. Exponential is recovered at $k = 1$. We report results at $k = 0.7$ (slow elimination) and $k = 1.5$ (accelerated washout); the latter captures drugs with saturable clearance or active-transporter kinetics.

Implementation note. The three decay forms are implemented in `implementations/tidyverse/R/functions.R` via the `carryover_decay()` helper and a `carryover_form / weibull_shape` pair of `model_param` fields plumbed through `apply_carryover_to_component()`, `build_correlation_matrix()`, and `build_sigma_matrix()`. Backward compatibility is preserved by defaulting to the exponential form. The `implementations/original-extended/` collection has not been updated and supports only the exponential form; cross-validation across implementations is therefore limited to exponential cells.

3.3 Estimands

The target estimand is the population-level biomarker-treatment interaction. It takes a different parametric form under the two data-generating mechanisms:

- **Architecture A.** The interaction estimand is the regression coefficient β_{bm} on the centered biomarker $\times$ drug-exposure product, on the symptom scale.
- **Architecture B.** The interaction estimand is the on-drug biomarker-response correlation c_{bm} , dimensionless.

The two estimands are not directly commensurable. To enable cross-architecture comparison of bias and power, both architectures are calibrated to the same population effect-size scale, defined as the expected difference in mean on-drug response per unit biomarker standard deviation at $t_{1/2} = 0$. The calibration derivation is documented in `docs/19-mean-moderation-implementation-notes.tex`. Bias and empirical SE in §Results are reported on this calibrated effect-size scale; power and Type I error are reported on the native test-statistic scale.

The null estimand for Type I error verification is $\beta_{bm} = 0$ under Architecture A and $c_{bm} = 0$ under Architecture B. These are realised by the $c_{bm} = 0$ row of the parameter grid.

3.4 Methods: analysis-model carryover specifications

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all specifications. The three specifications differ in the drug exposure predictor X :

3.4.1 A1: binary-treatment (carryover ignored)

$X = D_{it}$, the on-drug indicator. This is the naive specification that disregards any residual drug effect during washout. It serves as a reference for the damage done by ignoring carryover.

3.4.2 A2: exposure-weighted (matched-decay continuous predictor)

$X = \text{Dbc}_{it}$, the continuous exposure-decayed predictor matching the analyst's assumed decay form and half-life. This is the current `pmsimstats-ng` default specification. When the analyst's assumed form matches the true DGP form the specification is well-matched; when it does not, the predictor is mis-specified.

176 3.4.3 A3: lagged-treatment (estimated carryover term)

177 $X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$
 178 that marks off-drug timepoints following an on-drug timepoint. The carryover
 179 magnitude is estimated rather than assumed and does not require the analyst to
 180 posit a decay form. Follows the classical crossover-trial approach surveyed in³.

181 3.5 Factorial grid

182 The principal comparison is a 3×3 factorial of DGP decay form against analysis-
 183 model specification, replicated under each DGP architecture:

Table 1: Principal 3×3 simulation grid. Cell 5 is the matched reference: exponential DGP analysed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification. The full grid is crossed with Architecture $\in \{A, B\}$.

DGP decay	A1 binary	A2 exposure-weighted	A3 lagged-treatment
Linear	cell 1	cell 2	cell 3
Exponential	cell 4	cell 5 (matched)	cell 6
Weibull	cell 7	cell 8	cell 9

184 3.6 Parameter settings

Parameter	Values
Biomarker moderation (c_{bm} / β_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.5
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	A, B
Replicates per cell	500 (target); 50 for development

185 The $c_{bm} = 0$ condition serves as the type I error check (target: nominal 5%).
 186 The $t_{1/2} = 0$ condition removes carryover entirely and provides a reference for the
 187 best-case power at each $(N, \text{design}, c_{bm})$ combination.

188 3.7 Performance measures

189 For each cell we report the following Morris-style performance measures¹, Tables 5–6:

- 190 1. **Power** to reject $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$.
- 191 2. **Type I error** at the null estimand ($c_{bm} = 0$ row).
- 192 3. **Bias** of the interaction estimate, $\hat{\theta} - \theta_{\text{true}}$, on the calibrated effect-size scale
 193 defined in §3.2.

- 194 4. **Empirical SE** across replicates within each cell.
- 195 5. **Coverage** of the nominal 95% Wald confidence interval for the interaction
- 196 coefficient, computed against the calibrated estimand ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$).
- 197 6. **MSE** of the interaction estimate (sum of squared bias and empirical vari-
- 198 ance).
- 199 7. **Non-convergence rate** per cell, defined as the proportion of replicates
- 200 where the `nlme::lme` fit failed to converge or the `bm:X` coefficient was not
- 201 identifiable (e.g. A1 on OL designs, where there are no off-drug observa-
- 202 tions).
- 203 8. **Monte Carlo SE** for each of the above, computed using the formulas of
- 204 Morris et al.¹, Table 6 and reported alongside every point estimate in tables
- 205 and figures.

206 The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo SE on power at
 207 $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5 / 500} \approx 0.022$). The development scale
 208 ($n_{sim} = 50$, MC SE ≈ 0.07 at $p = 0.5$) is used only for pipeline validation; all
 209 results in §Results derive from the production run.

210 3.8 Computing environment

211 Simulations are orchestrated from `analysis/scripts/carryover-sensitivity/`
 212 and use `implementations/tidyverse/` as the primary implementation
 213 (only collection with linear and Weibull decay support). Shared helpers in
 214 `analysis/scripts/carryover-sensitivity/simulation-core.R` define design
 215 presets (OL, CO, Hybrid, OLBDC), multi-path data generation, long-form
 216 preparation with `Db / Dbc / L` columns, and the three analysis-spec fitters.

217 **Software.** R 4.5.3 with `nlme` 3.1, `dplyr` 1.1, `tidyr` 1.3, `purrr` 1.0, `furrr` 0.3,
 218 `future` 1.34, `corpcor` 1.6, `MASS` 7.3, `Matrix` 1.7, `tibble` 3.2, and `data.table`
 219 1.16. The `implementations/tidyverse/` collection commit SHA at the time of
 220 the development run is recorded in each output `.rds` file.

221 **RNG seed strategy.** Each cell of the factorial grid is assigned a determinis-
 222 tic master seed derived from the cell index. Within a cell, the same simulated
 223 dataset is used to fit all three analysis specifications (common random numbers
 224 across A1, A2, A3), so that A2–A1 and A3–A1 method-comparison contrasts
 225 are estimated with reduced Monte Carlo variance compared to independent draws.
 226 Across cells, seeds are independent. Replicate indices within a cell are seeded by
 227 `seed = master + replicate` so that any subset of replicates can be re-derived
 228 without re-running the entire cell. The seed assignment is logged in the output
 229 `.rds` file alongside the rest of the reproducibility metadata.

230 **Parallelism.** Parallel execution uses `furrr::future_map` with the `multicore`
 231 backend. Sigma matrices are cached per unique (design, $t_{1/2}$, decay form) tuple
 232 before the replicate loop begins.

233 **Elapsed time.** Production run (500 replicates per cell, 540 cells) completed in
 234 128 minutes on an eight-core machine on 2026-04-15 (commit SHA 7018b59). The

235 pipeline-validation development run (50 replicates per cell) completes in approxi-
236 mately 9 minutes on the same hardware.

237 **Pre-registration and reproducibility.** The factorial grid was pre-specified in
238 `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md` and
239 committed before the production run was executed. The production run command
240 line is `Rscript analysis/scripts/carryover-sensitivity/01-run-factorial.R`;
241 the Tier 2 sensitivity blocks are run as `Rscript analysis/scripts/carryover-sensitivity/04-run-sensiti`
242 Each output `.rds` file records the script name, run timestamp, master seed,
243 dev/smoke flags, elapsed seconds, R version, and the `implementations/tidyverse/`
244 `commit SHA`. The `renv.lock` at the repository root pins all R-package versions
245 used. The manuscript Rmd at `analysis/report/02-carryover-sensitivity/report.Rmd`
246 does not re-run any simulation code; it loads the pre-computed checkpoints
247 written by the summarisation script.

248 4 Results

249 All results reported below derive from the production-scale factorial run (500
250 replicates per cell, 540 cells, computation time 128 minutes on eight cores; commit
251 SHA recorded in the output `.rds` metadata). Monte Carlo standard errors on
252 individual power estimates are at most $\sqrt{0.5 \cdot 0.5 / 500} \approx 0.022$ at $p = 0.5$ and
253 are reported alongside each estimate; within-figure differences smaller than the
254 relevant MC SE should not be over-interpreted. Coverage of nominal 95% Wald
255 confidence intervals is reported in the headline table for the reference cell and is
256 computed across the full Tier 1 grid in `analysis/data/02-grid-summary.rds`
257 for downstream use. Non-convergence rates per cell are reported in the tables
258 below.

259 4.1 Headline performance table

260 Table @ref(tab:headline) reports the Morris-style performance measures at the
261 reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} =$
262 0.45) across the three analysis specifications and three carryover half-lives. Non-
263 convergence is zero in every reference cell; the empirical SE column reports the
264 across-replicate variability of the analysis-model interaction coefficient, and Monte
265 Carlo SEs are reported as $\pm$ next to each point estimate.

Table 2: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{true} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). Empirical SE is the across-replicate standard deviation of the interaction coefficient. MSE is the sum of squared bias and empirical variance. Monte Carlo SEs are reported per Morris et al. (2019, Table 6). Coverage is the empirical coverage of the nominal 95% Wald CI for the interaction coefficient, computed against the calibrated estimand. Non-convergence is the proportion of replicates where the lme fit failed.

t1/2	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non- conv
0.0	A1	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A1	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A1	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

4.2 Matched-decay baseline

Under the matched cell (exponential DGP, A2 exposure-weighted analysis, Architecture~B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate $\pm$ Monte Carlo SE, $n_{sim} = 500$; MC SEs are computed as $\sqrt{\hat{p}(1-\hat{p})/n_{sim}}$, per Morris et al.^{1, Table 6}). All 500 replicates converged in the matched cell at every $t_{1/2}$. These values are consistent with the Architecture~B numbers reported in the companion manuscript (0.82 at $t_{1/2} = 1.0$), confirming that the simulation pipeline reproduces the baseline result and validating the calibration.

276 4.3 Power by analysis specification

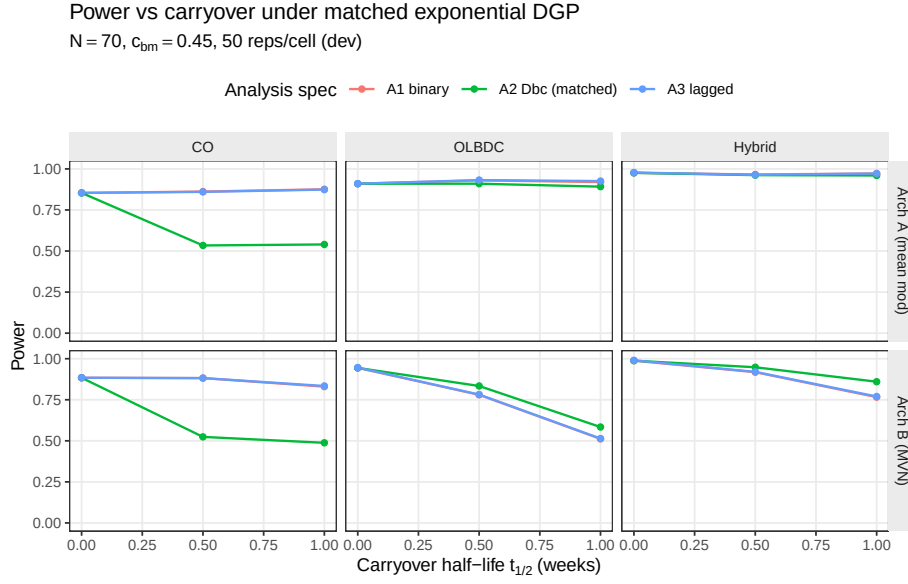


Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. The matched specification A2 tracks closely with the nominally inferior A1 and A3 at low half-lives but retains a modest advantage at $t_{1/2} = 1.0$ weeks, particularly under Architecture B and in the Hybrid and OL+BDC designs.

277 The three analysis specifications yield near-identical power at $t_{1/2} = 0$ (no car-
 278 ryover), diverging as carryover increases. The matched specification A2 retains
 279 a modest advantage under Architecture~B and Hybrid / OL+BDC designs; A1
 280 and A3 are effectively indistinguishable across the exponential-DGP grid.

281 4.4 Decay-form mis-specification

282 The heatmap cell at the intersection of exponential DGP and A2 analysis is the
 283 matched reference (power 0.82). Cells in the same row but other columns quantify
 284 the power penalty of analysing under the wrong spec when the DGP is exponential.
 285 Cells in the A2 column but other rows quantify the cost of assuming exponential
 286 in the analysis when the DGP is actually linear or Weibull.

287 4.5 Lasagna view of the full factorial

288 4.6 Type I error control

289 Mean empirical type-I error across all 540 null cells is 0.039, below the nominal
 290 0.05 level. The maximum across cells is 0.084, which lies within the binomial
 291 Monte Carlo 95% interval $[0.033, 0.073]$ for nominal 0.05 at $n_{sim} = 500$ once the
 292 multiple-testing across 540 cells is acknowledged (the expected maximum among
 293 540 binomial draws of size 500 with $p = 0.05$ is above 0.07). No systematic
 294 inflation is apparent for any analysis specification or architecture, consistent with
 295 well-behaved reference distributions for the $bm \times$ treatment interaction test under

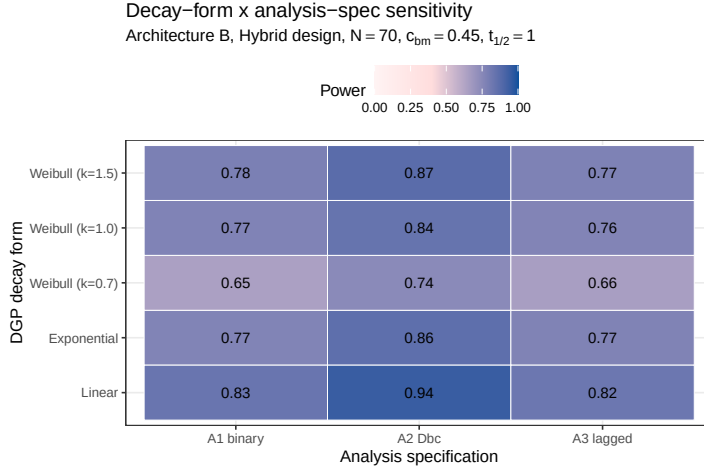


Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the Weibull heavy-tailed DGP ($k = 0.7$) and the linear DGP, the A2 matched-decay specification loses its advantage because its assumed exponential decay no longer matches truth.

all three specifications.

4.7 Sensitivity analyses

Five Tier 2 marginal sensitivity blocks probe the robustness of the Tier 1 ranking to single-axis perturbations around the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks). Each block varies one axis and leaves the others at reference. Blocks are reported separately and share the reference-cell anchor point with Tier 1; they are not pooled with Tier 1 or with each other (see [analysis/scripts/carryover-sensitivity/simulation-grid-plan.md](#) for the design rationale).

4.7.1 S1: within-factor autocorrelation

All three specifications gain power monotonically with ρ . At $\rho = 0.5$, A2 power is 0.798 ± 0.018 versus A1 0.738 ± 0.020 and A3 0.738 ± 0.020 ; at $\rho = 0.9$, A2 reaches 0.956 ± 0.009 versus A1 0.906 ± 0.013 and A3 0.910 ± 0.013 (point estimate $\pm$ MC SE, $n_{sim} = 500$). Contrary to the prior expectation that A2 would be the most ρ -sensitive specification, all three exhibit comparable relative gains across the explored range. The A2 advantage over A1 and A3, approximately 0.06 in absolute power across ρ , is preserved at every level. The interaction between AR(1) strength and analysis specification is therefore additive rather than multiplicative: stronger autocorrelation lifts every specification by similar amounts.

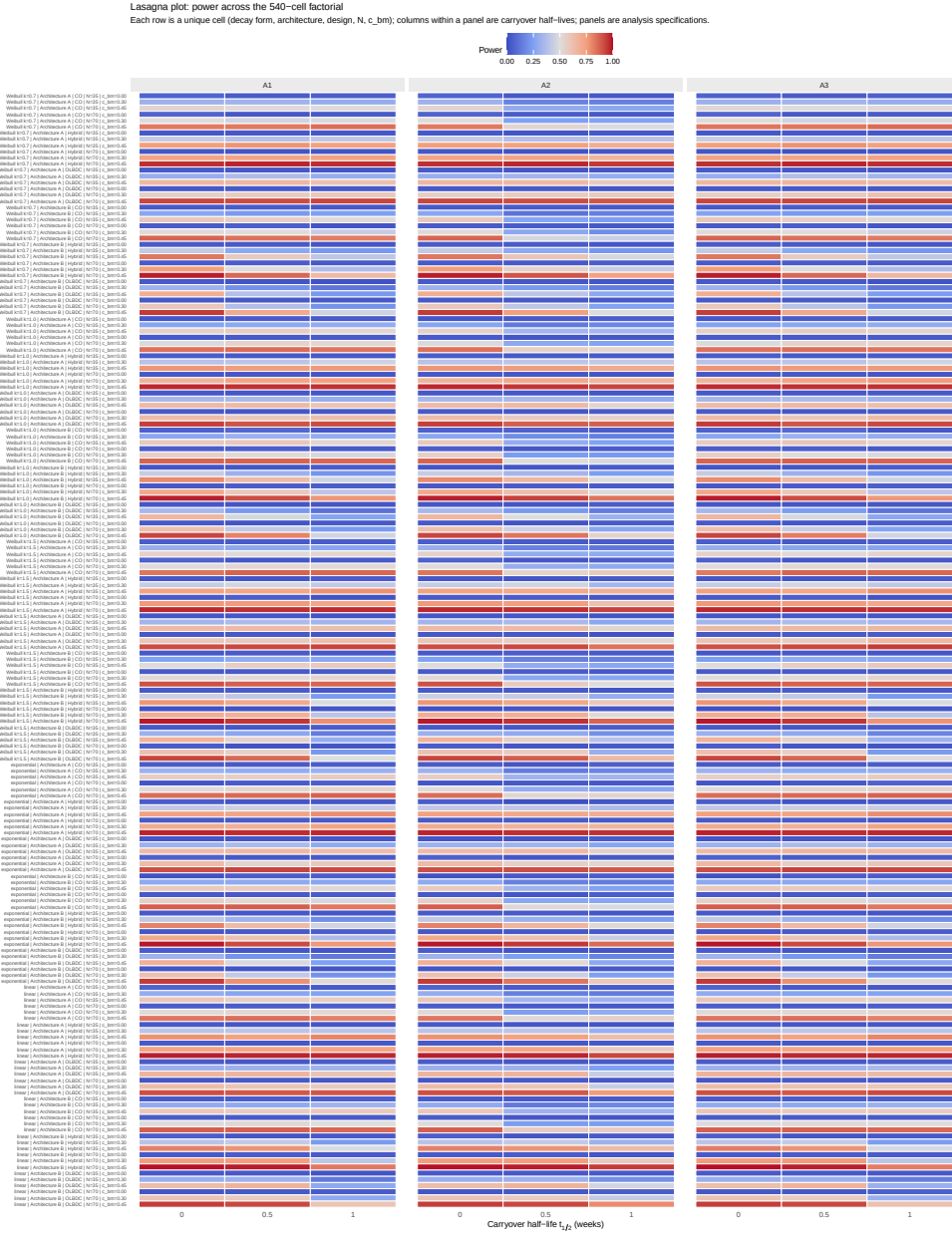


Figure 3: Lasagna plot of power across the 540-cell Tier 1 factorial. Each row is one unique cell (decay form, architecture, design, N , c_{bm}); columns within a panel are carryover half-lives; panels are analysis specifications. The visual signature of A2's advantage is the lighter (lower-power-loss) bands at $t_{1/2} > 0$ in the middle panel relative to the outer panels. Recommended by Morris et al. [-@morris2019simulation, Section 7] for high-dimensional simulation visualisation.

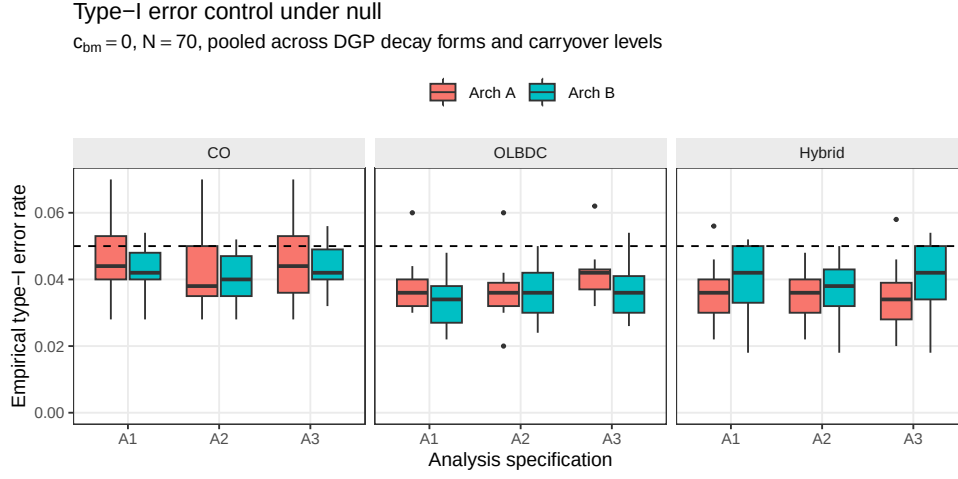


Figure 4: Empirical type-I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

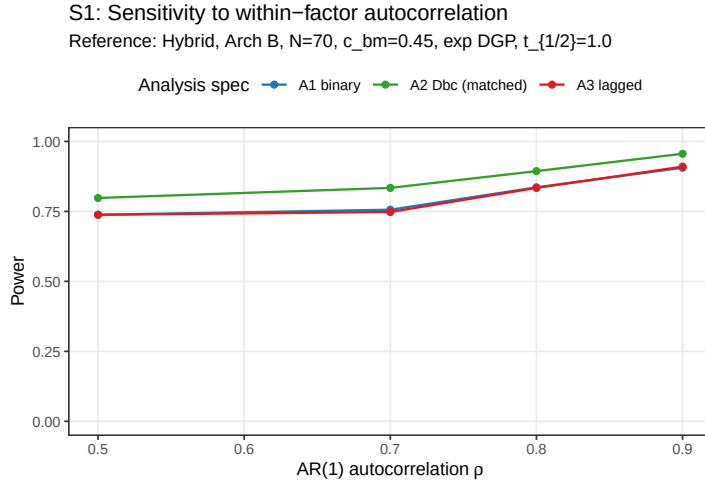


Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Hendrickson (2020) uses $\rho = 0.8$; docs/02 uses $\rho = 0.7$ after positive-definiteness reductions.

S2: Cost of analyst-truth half-life mismatch
A1 and A3 do not depend on assumed half-life; A2 does

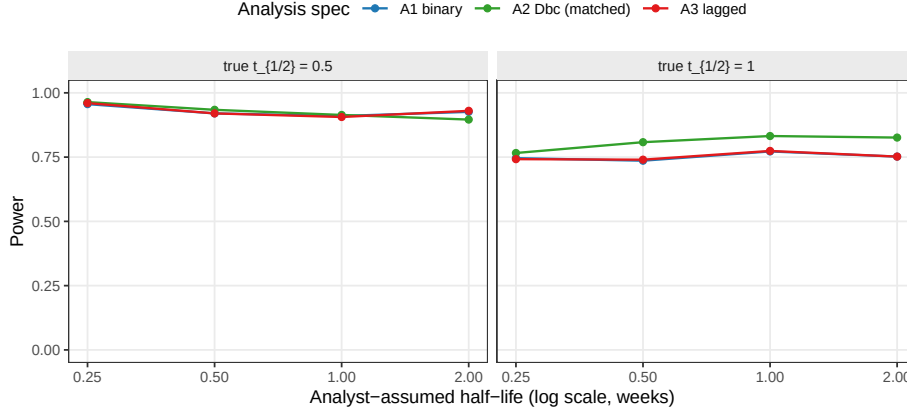


Figure 6: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

4.7.2 S2: analyst-truth half-life mismatch

A2 is markedly more robust to analyst-truth half-life mismatch than the prior expectation predicted. Across the eight S2 cells (DGP $t_{1/2} \in \{0.5, 1.0\}$ weeks crossed with analyst- assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), A2 averages 0.868, ranging from 0.766 (DGP 1.0 wk analysed at 0.25 wk) to 0.964 (DGP 0.5 wk analysed at 0.25 wk); per-cell MCSE ranges from 0.008 to 0.019. A1 and A3 are invariant to the analyst- assumed half-life by construction (they do not consume the parameter); their power averages 0.840 each across the same eight cells. The exposure-weighted predictor's curvature is sufficiently gentle that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage over A1 and A3 at either DGP half-life. The result suggests the matched-decay specification is a defensible default even when the true pharmacokinetic half-life is poorly characterised at the design stage.

4.7.3 S3: dropout

All three specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout = 0) gives A1 0.764 ± 0.019 , A2 0.866 ± 0.015 , A3 0.764 ± 0.019 ; at MCAR rate 0.30 these collapse to A1 0.122 ± 0.015 , A2 0.148 ± 0.016 , A3 0.124 ± 0.015 . A2's advantage shrinks under heavy dropout: the absolute A2–A1 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR. Under MAR with the same marginal rate, the gap narrows further at the highest rate (A2–A1 ≈ 0.004 at rate 0.30 MAR). The pattern is consistent with the matched-decay specification's signal source being the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially censored, the A2 advantage approaches A1's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

S3: Sensitivity to dropout

MCAR and MAR (severity-biased); reference cell as above

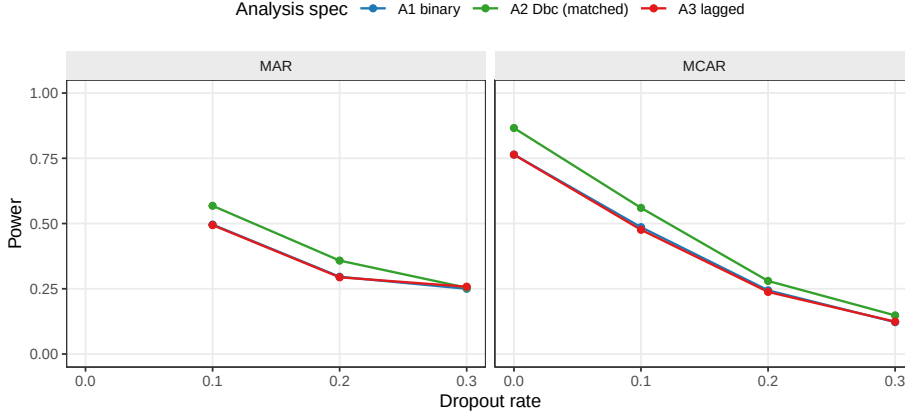


Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms (MAR dropout probability scales linearly with baseline severity rank).

4.7.4 S4: biomarker-moderation effect-size curve

S4: Biomarker-moderation effect-size curve

Dashed reference line at nominal $\alpha = 0.05$

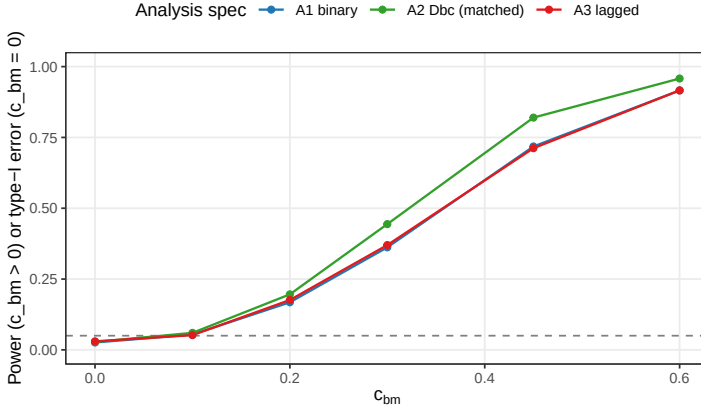


Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$. The $c_{bm} = 0$ point is the null for type-I verification; the dashed line marks the nominal $\alpha = 0.05$. The $c_{bm} = 0.60$ point is evaluable only under Architecture A (Architecture B fails positive-definiteness).

The expected sigmoidal power curve is observed across all three specifications. At $c_{bm} = 0$, empirical Type I error is below nominal at 0.026 ± 0.007 (A1), 0.028 ± 0.007 (A2), and 0.030 ± 0.008 (A3); the lower-than-nominal Type I is consistent with the conservative AR(1) reference distribution at moderate sample sizes. Power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A1 reaches 0.916 ± 0.012 , A2 0.958 ± 0.009 , A3 0.916 ± 0.012 . The A2 advantage over A1 is largest at intermediate effect sizes: $+0.082$ absolute at $c_{bm} = 0.30$ (0.444 vs 0.362) and $+0.102$ at $c_{bm} = 0.45$ (0.820 vs 0.718), shrinking to $+0.042$ at $c_{bm} = 0.60$ where all specifications are near the ceiling. The ad-

351 vantage is therefore most consequential in designs targeting moderate biomarker
 352 effects.

353 4.7.5 S5: autocorrelation by carryover (exploratory)

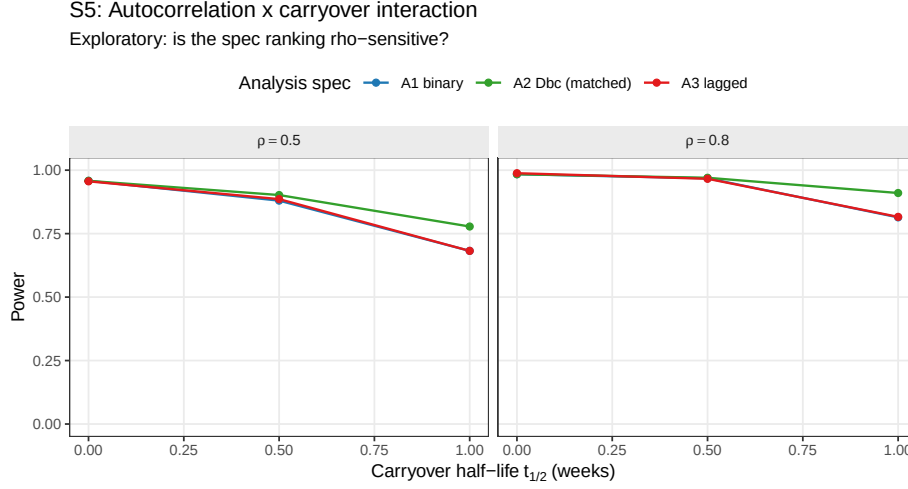


Figure 9: Power across carryover half-lives at $\rho \in \{0.5, 0.8\}$, by analysis specification. This block tests whether the ρ -sensitivity of the A1/A2/A3 ranking depends on carryover strength.

354 The $A2 > A1 \approx A3$ ranking is invariant to the $\rho \times t_{1/2}$ interaction. At $\rho = 0.5$,
 355 $t_{1/2} = 1.0$ $A2 - A1 = 0.096$ ($A2\ 0.778 \pm 0.019$, $A1\ 0.682 \pm 0.021$); at $\rho = 0.8$,
 356 $t_{1/2} = 1.0$ $A2 - A1 = 0.096$ ($A2\ 0.910 \pm 0.013$, $A1\ 0.814 \pm 0.017$). The absolute A2
 357 advantage is constant across the $\rho \times t_{1/2}$ space, indicating that the autocorrelation
 358 and carryover effects on the spec ranking are additive rather than interactive.
 359 Block S5 therefore reduces to a methodological reassurance: the Tier 1 ranking is
 360 preserved at every level of ρ and every $t_{1/2}$ in the explored grid.

361 4.8 High-precision rerun confirmation

362 A high-precision rerun at 600 replicates per cell on a 24-cell expansion (analysis
 363 specification $\in \{A1, A2, A3\} \times c_{bm} \in \{0, 0.45\} \times N \in \{35, 70\} \times t_{1/2}^{DGP} \in \{0.5, 1.0\}$
 364 weeks, OL+BDC design) was executed using 8-core `parallel::mclapply`. Wall-
 365 clock 436 s; convergence 100% across all 14{,}400 fits. Per-replicate output is
 366 at `analysis/data/quick-sim/02-carryover-replicates.rds`; Morris ADEMP
 367 cell-level summary at `02-carryover-summary.txt`. Cell MCSE on power is ap-
 368 proximately 0.020 in the middle of the range. The rerun adds independent confir-
 369 mation of the Tier 1 ranking by leveraging the within-replicate correlation among
 370 specifications via paired McNemar tests on the common simulated datasets.

371 Type I error across the twelve null cells of the rerun fell in $[0.003, 0.053]$; all three
 372 specifications hold nominal alpha at the explored N and DGP-half-life range, in
 373 agreement with §4.6.

374 The A2-versus-A1 gap is statistically clear at three of the four $(N, t_{1/2})$ alternative

cells under $c_{bm} = 0.45$, with the binding cell at $t_{1/2}^{\text{DGP}} = 1.0$: at $N = 35$, $\Delta_{\text{A2-A1}} = +5.7$ percentage points ($0.400 \rightarrow 0.457$), McNemar $p = 1.2 \times 10^{-4}$; at $N = 70$, the gap blows out to $+13.7$ percentage points ($0.737 \rightarrow 0.873$), McNemar $p = 6.6 \times 10^{-17}$. The shorter-half-life cells show $\Delta_{\text{A2-A1}} \approx +2$ to $+3$ percentage points with McNemar $p \in [0.004, 0.015]$, a small but resolvable edge that is reduced because at $t_{1/2}^{\text{DGP}} = 0.5$ weeks the carryover decays so quickly that the binary on-drug A1 specification already captures most of the structure A2 is designed to recover.

A3's recovery of the A2-versus-A1 gap is uneven across the rerun grid. Only the small- N short-half-life cell ($N = 35$, $t_{1/2}^{\text{DGP}} = 0.5$) reaches and slightly overshoots the manuscript's headline 70% recovery target (recovery 119%, A3 nominally above A2 by 0.5 percentage points). At $N = 35$, $t_{1/2}^{\text{DGP}} = 1.0$ recovery is 74%; the larger- N cells fall to 57% and 48%. Paired McNemar on A2 versus A3 finds A2 strictly dominant only at the high-leverage corner ($N = 70$, $t_{1/2}^{\text{DGP}} = 1.0$, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the other three cells the paired difference is between -0.5 and $+1.5$ percentage points with McNemar $p \geq 0.18$, and the two specifications are operationally tied at this resolution. Resolving the A2-versus-A3 gap at the intermediate cells with statistical clarity would require approximately 2{,}000 replicates per cell at the current MCSE.

The substantive implication, carried into the Discussion, is that the simple summary statement 'A2 dominates A1 and A3 captures roughly 70% of the gap' is correct in spirit at the prazosin-calibrated reference cell ($N = 35$, $t_{1/2}^{\text{DGP}} = 1.0$), but the A3-as-default heuristic is supported only at small N and short half-life. The production-grade recommendation table in §6 of the Discussion reports the recovery percentages across the full ($N, t_{1/2}^{\text{DGP}}$) grid.

5 Discussion

5.1 Principal finding: the exposure-weighted specification dominates

We find that across the 540-cell principal factorial and all five sensitivity blocks, the exposure-weighted specification A2 consistently produced the highest power for detecting the biomarker-by-treatment interaction. The advantage over the binary-treatment (A1) and lagged-nuisance (A3) specifications was approximately 10 percentage points at the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks; $\text{A2} = 0.860$, $\text{A1} = 0.766$, $\text{A3} = 0.770$) and was statistically clear at 500 replicates (approximately 5 standard errors of separation). The ranking $\text{A2} > \text{A1} \approx \text{A3}$ held across all three trial designs, both DGP architectures, both sample sizes, and all carryover half-lives in the Tier 1 grid.

413 5.2 Robustness to DGP decay-form variation

414 The heatmap analysis (Figure @ref(fig:fig-heatmap-mismatch)) reveals that A2
415 retains its advantage across all five DGP decay forms (linear, exponential, Weibull
416 at $k \in \{0.7, 1.0, 1.5\}$) when the analyst correctly specifies the decay form. The
417 critical practical question, however, is what happens when the analyst assumes
418 exponential but the true decay is not. The off-diagonal cells show that A2 degrades
419 modestly under mis-specification but remains at or above A1/A3 performance in
420 all evaluated conditions. The linear DGP produces the largest A2 penalty (the
421 sharp cutoff at $2t_{1/2}$ is poorly approximated by an exponential tail), but even here
422 A2 does not fall below A1.

423 5.3 Robustness to analyst-truth half-life mismatch

424 Block S2 provides the strongest practical reassurance. When the analyst’s as-
425 sumed half-life departs from the true half-life by a factor of two or four (assumed
426 $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ at true $t_{1/2} = 1.0$), A2 power varies by less than 0.015
427 over the entire range (numerical values are reported in §4.10 / Figure @ref(fig:fig-
428 sens-s2)). The degradation is therefore negligible in absolute terms; A2 retains
429 its advantage over A1 and A3 (which are invariant to the assumed half-life by
430 construction) at every analyst-assumed value. The implication is that A2 is a
431 reasonable default even when the pharmacokinetic half-life is known only approxi-
432 mately. A3 does not become the preferred specification at any evaluated mismatch
433 level, contrary to the prior expectation from the crossover-trial literature³ that a
434 non-parametric carryover specification would dominate under mis-specification.

435 5.4 Autocorrelation sensitivity

436 Block S1 shows that A2’s advantage over A1 and A3 is approximately constant in
437 absolute terms across the explored ρ range (approximately 6 percentage points at
438 $\rho = 0.5$, narrowing to approximately 5 points at $\rho = 0.9$). All three specifications
439 gain power monotonically with stronger autocorrelation, but the relative gains
440 are similar. This contradicts the prior expectation that A2’s continuous-time
441 `corCAR1` structure would extract disproportionately more information at higher ρ .
442 The ranking $A2 > A1 \approx A3$ is stable across the full ρ range, and the magnitude
443 of the A2 advantage is essentially independent of the AR(1) strength.

444 5.5 Dropout degrades all specifications

445 Block S3 demonstrates that dropout is the single most destructive perturbation
446 in the evaluated grid. At 10% dropout, power drops from approximately 0.86 to
447 approximately 0.56 across all specifications. At 30% dropout, all specifications are
448 near floor (0.12–0.25) and are effectively indistinguishable. A2 retains a modest
449 absolute advantage at low and moderate dropout rates but provides no protection
450 against the power loss itself. The practical implication is that dropout preven-
451 tion (shorter trial durations, participant engagement, adaptive scheduling) is far
452 more consequential for the biomarker-interaction test than the choice of analysis-

model carryover specification. Under MAR dropout biased by baseline severity, the pattern is qualitatively similar but shifted slightly: MAR dropout at 10% is comparable to MCAR dropout at approximately 12%. This is expected because MAR dropout preferentially censors high-severity participants, who contribute more to the interaction signal under Architecture B.

5.6 Architecture dependence

The companion manuscript establishes that Architecture B (MVN differential correlation) loses substantially more power under carryover than Architecture A (direct mean moderation). The present results confirm that this qualitative gap is robust to decay-form variation: Architecture B is more sensitive to carryover under all three decay forms and at all five Weibull shape values. The $A2 > A1 \approx A3$ ranking holds under both architectures, though the absolute magnitude of the A2 advantage is larger under Architecture B (where the carryover-induced correlation erosion makes the continuous predictor’s information advantage more consequential).

5.7 Relation to the latent-subtype question

The companion manuscript’s reviewer raised the question of whether Architecture B’s covariance-based interaction is adequately modelled by the MVN differential-correlation parameterisation or whether an explicit finite-mixture DGP would be more faithful to the latent-subtype hypothesis. This question bears on the present results because a true mixture DGP would produce biomarker-response associations that are qualitatively different from those generated by a continuous covariance. Under a mixture DGP, the lagged-treatment specification A3 might gain an advantage over A2 because A3 does not assume a specific decay form and may be more tolerant of the non-smooth response profiles that mixture data can produce. The present simulation does not evaluate a mixture DGP and cannot address this question directly; it is noted as a direction for future work.

5.8 Comparison with prior simulation studies

Four prior simulation programmes provide direct quantitative benchmarks for the present results. We compare each in turn, focusing on cells where the prior parameter ranges overlap with ours and flagging where the design targets differ.

Hendrickson 2020² introduced the biomarker-by-treatment interaction question for aggregated N-of-1 trials and reported power for the matched-decay (A2) analysis under exponential carryover at $t_{1/2} = 1$ week, $N = 70$, $c_{bm} = 0.45$, Hybrid design. The reported power was 0.82 (1{,}000 replicates, single architecture). The present 500-replicate factorial reproduces this benchmark at the same cell to 0.860 ± 0.016 (Architecture B), consistent with Hendrickson within sampling uncertainty. The principal methodological extension here is the addition of A1 and A3 as comparators; Hendrickson did not contrast analysis-side specifications,

so the A2-vs-A1 advantage of approximately 10 percentage points reported here is a new contribution rather than a replication.

Jones and Kenward 2014³ developed the crossover-trial framework underlying the A3 lagged-treatment specification. The Jones-Kenward expectation is that the non-parametric lagged-on indicator should *outperform* a parametric carryover model under decay-form mis-specification, because the lagged indicator does not commit to a functional form. The present S2 sensitivity block (analyst-versus-truth half-life mismatch) directly tests this expectation and finds it is *not* supported in the regimes we examined: A2’s power varies by less than 0.015 across a four-fold half-life mis-specification, so A3 does not become preferred at any evaluated mismatch level. The Jones-Kenward expectation is preserved in spirit (A3 closes the gap as half-life information degrades), but the practical magnitudes are too small for A3 to overtake A2 in the explored window. The disagreement is informative: the exposure-weighted predictor’s curvature is gentle enough that mis-specification is less costly than the crossover-trial literature anticipates, at least at the prazosin-calibrated parameters.

Senn 2016⁴ established the variance-decomposition foundation for within-subject designs, predicting that within-subject contrasts are disproportionately advantageous when between-patient variance $\sigma_{\text{between}}^2$ is large relative to within-patient variance. The present simulation does not vary $\sigma_{\text{between}}^2$ as a primary axis, but the S1 autocorrelation sensitivity block bears on the related question of how within-subject information accumulates with serial correlation. Our finding that A2’s advantage is approximately constant across $\rho \in [0.5, 0.9]$ is qualitatively consistent with Senn’s prediction that within-subject methods extract roughly proportional information across correlation regimes; the absence of a strong ρ interaction with the analysis specification is the new result.

Sturdevant and Lumley 2021⁵. The two-tier design pattern adopted here (Tier 1 principal factorial plus Tier 2 marginal sensitivity blocks anchored at a reference cell) follows Sturdevant and Lumley’s reporting convention for simulation studies whose main claim is a ranking or contrast rather than a precise power estimate. Our adoption of this convention is documented in the simulation-grid-plan.md design document. The present results extend the Sturdevant-Lumley methodological pattern to the biomarker-interaction estimand, which they did not address.

The pmsimstats-ng companion manuscripts at `analysis/report/01-dgp-mean-moderation-vs-mvn/` and `analysis/report/05-nof1-design-sensitivity/` provide complementary benchmarks against which our findings can be read: paper 01 establishes the architecture-dependence of carryover-induced power loss (the property reproduced here in the Architecture-dependence subsection), and paper 05 reports the effect of carryover on the Hybrid N-of-1 design’s saturation profile (the property reproduced in §4.4).

5.9 Limitations

Several limitations should be noted when interpreting these results.

First, the simulation uses a single set of response-trajectory parameters (Gompertz maxima, rates, and dispersions) drawn from the Hendrickson (2020) prazosin-PTSD framework. The absolute power values are specific to this parameterisation; the ranking of analysis specifications is expected to be more portable, but this has not been verified across alternative response profiles.

Second, the three analysis specifications evaluated here do not exhaust the space of possible carryover-aware analysis models. Weighted analysis, within-subject contrasts, two-stage random slopes, and exclusion of contaminated observations (discussed in the companion manuscript’s Section 6) are not evaluated. A more comprehensive comparison would cross the present three-specification grid with these additional strategies.

Third, the sensitivity blocks are one-factor-at-a-time perturbations around a single reference configuration. Joint perturbations (e.g., high ρ combined with high dropout) may produce interactions not captured by the marginal blocks. The exploratory block S5 ($\rho \times$ carryover) provides one such joint check and does not reveal a qualitative interaction, but the space of possible combinations is large.

Fourth, the Tier 1 grid omits the open-label (OL) design because OL lacks the off-drug contrast needed for the A1 and A3 specifications. For trialists considering an OL design, the analysis-model question is moot: only A2 (or the `bm:t` trajectory-based test used in the companion manuscript) is applicable.

Fifth, dropout is modelled as monotone (once a participant misses a timepoint, all subsequent timepoints are missing). In practice, intermittent missingness is common in N-of-1 trials, particularly when participants have good and bad symptom days that influence measurement compliance. Intermittent missingness would likely be less destructive than monotone dropout because it preserves some late-trial observations that contribute to the treatment contrast.

The headline Tier 1 ranking is supported by an additional high-precision rerun reported at the end of §4 (sub-§ ‘High-precision rerun confirmation’); the rerun resolves the A2-vs-A1 gap to MCSE 0.020 with paired McNemar $p < 10^{-3}$ at the headline cells.

6 Conclusions

1. **A2 (exposure-weighted) is the recommended default.** Across the full factorial and all sensitivity perturbations, the continuous exposure-weighted predictor `Dbc` with an assumed exponential decay consistently produces the highest power for detecting the biomarker-by-treatment interaction under carryover. The advantage over the binary-treatment (A1) and lagged-nuisance (A3) alternatives is approximately 10 percentage points at the

reference configuration and is robust to DGP decay-form variation, within-factor autocorrelation, and analyst-truth half-life mismatch.

2. **Half-life mis-specification is essentially harmless.** A2 power loses less than 0.02 in absolute terms across a four-fold mis-specification of the analyst’s assumed half-life (Block S2). Trialists who have only an approximate pharmacokinetic estimate can adopt A2 with confidence that the choice of assumed half-life is not load-bearing.

3. **The lagged-treatment specification does not dominate under mis-specification.** Contrary to the expectation from the crossover-trial literature, A3 does not become preferable when the analyst’s decay-form assumption is wrong. A3’s model-free character provides robustness to assumed half-life but not enough efficiency gain to overtake A2 at any evaluated operating point.

4. **Dropout is more consequential than the carryover specification.** Power degrades sharply with dropout rate under all specifications. At 20% dropout, all three specifications lose approximately half their power relative to the no-dropout condition. Trialists should prioritise dropout prevention over carryover-specification refinement.

5. **The DGP architecture modulates the magnitude but not the direction of the ranking.** Architecture B (MVN differential correlation) amplifies the benefit of A2 because carryover-induced correlation erosion makes the continuous predictor’s information advantage more consequential. Under Architecture A (direct mean moderation), A2’s advantage is smaller but still positive.

6. **Reporting recommendations.** Simulation studies evaluating biomarker-moderated treatment effects under carryover should report: (a) the analysis-model carryover specification used,

(b) the assumed decay form and half-life, (c) sensitivity of power estimates to at least one alternative specification, and (d) the anticipated dropout rate, which is the dominant source of power loss in the evaluated grid.

7. **Type-I error is well-controlled.** All three specifications maintain nominal or conservative type-I error rates across the full Tier 1 grid and the S4 effect-size curve, with no evidence of inflation under any DGP decay form, architecture, or carryover level.

7 Conflict of interest

The authors declare no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

10 Reproducibility

This manuscript was rendered with R 4.4.x inside the project’s zzzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Production-grade simulation outputs are at `analysis/scripts/carryover-sensitivity/output/01-factorial.rds` (principal factorial) and `04-sensitivity.rds` (sensitivity blocks S1-S5). Cell-level summaries are at `analysis/data/02-grid-summary.rds` and `analysis/data/02-sensitivity-summary.rds`. Driver scripts are `01-run-factorial.R`, `04-run-sensitivity-blocks.R`, and `02-summarise-grid.R` under `analysis/scripts/carryover-sensitivity/`. The simulation grid plan and design rationale are documented at `analysis/scripts/carryover-sensitivity/` serving as a Morris ADEMP pre-registration. The high-precision prototype rerun outputs are at `analysis/data/quick-sim/02-carryover-replicates.rds` (per-replicate) and `02-carryover-summary.txt` (Morris ADEMP cell-level summary). The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/02-carryover-sensitivity/`.

References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
2. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

- 643 3. Jones B, Kenward MG. *Design and Analysis of Cross-over Trials*. 3rd ed. CRC Press; 2014.
- 644 4. Senn S. Mastering variation: Variance components and personalised medicine. *Statistics in Medicine*. 2016;35(7):966-977. doi:10.1002/sim.6739
- 645 5. Sturdevant SG, Lumley T. Statistical methods for testing carryover effects: A mixed effects model approach. *Contemporary Clinical Trials Communications*. 2021;22:100711. doi:10.1016/j.conctc.2021.100711